

Krishna Koumudi Koravi

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SUMMARY

Python engineer with cross-cutting strength in AI/ML, data engineering, and backend development. Built SID, an open-source geospatial AI platform for the Järfälla municipality: data pipelines, fine-tuning AI foundation models, and REST APIs end to end. Built a LangGraph RAG agent that answers questions over a 364-page rulebook with page citations, scoring 12/12 on a custom eval set. Daily user of agentic AI tooling (Claude Code). Background in application security testing adds a rigorous eye to code quality.

SKILLS

- **Python & Software Engineering:** Python, OOP, Java, ASP.NET, Git/GitHub, GitLab, Docker, CI/CD, Linux
- **AI / ML Engineering:** TensorFlow, Scikit-learn, Pandas, NumPy, prompt engineering, LLM-based summarization, model fine-tuning
- **LLM Applications, RAG & Agents:** RAG, LangGraph, LangChain, tool-calling agents, ChromaDB, vector search, embeddings (sentence-transformers), LLM evaluation, GPT-4 via Azure AI Foundry, Groq, self-hosted open-source LLMs (Llama via OpenClaw), Streamlit
- **AI-Assisted Development:** Claude Code (daily use)
- **Data Engineering:** Data pipelines, ETL, MongoDB, SQLite, SQL, Azure Cosmos DB
- **Geospatial & Remote Sensing:** Google Earth Engine, Sentinel-2/3, Landsat, Dynamic World, IBM TerraMind, GIS
- **Backend & APIs:** FastAPI, Django, Flask, REST API development
- **Security (background):** Static/dynamic analysis, Ghidra, malware analysis

EXPERIENCE

AI Engineer Intern — Savantic AI Lab, Stockholm

Jun 2026 – Sep 2026 (completed)

- Developed SID (Satellite Intelligence for DASH): geospatial analysis features and backend services in Python and FastAPI, exposed through REST APIs for DASH, an open-source AI/GIS urban-planning platform for the Järfälla municipality
- Built a data pipeline to fetch and filter satellite imagery by lowest cloud cover for downstream analysis
- Developed a heat-island detection layer using Sentinel-3 SLSTR thermal imagery, with urban heat intensity coloring and integration of SMHI air-temperature data
- Validated and integrated Dynamic World (Google/WRI's pretrained land-cover model) via Google Earth Engine against 100 Järfälla Sentinel-2 scenes, producing a 10m-resolution land-cover overlay layer in SID
- Fine-tuned IBM TerraMind, a geospatial AI foundation model, on Sweden's National Land Cover (NMD) dataset, building the data pipeline, class mapping, training configuration, and evaluation workflow end to end
- Evaluated alternative land-cover sources including ESA WorldCover, and validated model outputs through accuracy analysis and testing
- Built an AI chat feature for SID that summarizes and surfaces insights from geospatial data layers in natural language, using GPT-4 via Azure AI Foundry
- Set up Docker-based CI/CD pipelines on GitLab for the SID/DASH codebase, alongside GitHub source control with environment-file and credential safeguards

Software Developer Analyst — Accenture, Hyderabad, India

Nov 2020 – Jan 2023

- Developed and optimized secure, high-performance Android applications in Java, applying secure coding practices against insecure data storage, hardcoded credentials, and improper API handling
- Conducted static and dynamic analysis of Android applications, including code reviews and validation against best-practice standards
- Used Git for version control, defect tracking, and cross-team collaboration
- Investigated, reproduced, and debugged issues using Android Studio and Logcat; reverse-engineered a heavily obfuscated application, uncovering critical vulnerabilities and earning recognition for analytical rigor

EDUCATION

- **Diploma, Python AI/ML Developer** — Lexicon IT Proffs, Stockholm (Nov 2025 – May 2026). Full-time, 40 hrs/week: Python & OOP, Django, MongoDB & SQLite, Docker/Git/GitHub, ML/deep-learning foundations, applied AI including prompt engineering and Azure-based cloud data analysis
- **M.Sc. Computer Science** — Osmania University (2017 – 2019)
- **B.Sc. Computer Science** — Osmania University (2014 – 2017)

PROJECTS

- **The Rules Lawyer, RAG Agent** (github.com/krish26/RAG_Agent): LangGraph tool-calling agent answering questions over the 364-page D&D rules PDF with page citations. Structure-aware parsing and chunking (3,618 chunks, tables kept intact), local embeddings in ChromaDB, and table-lookup tools; raised accuracy from 8/12 (classic RAG) to 12/12 on a custom eval set, including correct refusals
- **Flask REST API with Azure Cosmos DB**: REST API in Flask backed by Azure Cosmos DB, deployed to Azure App Service
- **Swedish Job Market Analytics Pipeline**: end-to-end Python ETL pipeline against Arbetsförmedlingen's Jobtech API
- **Phishing URL Detection Model**: Random Forest and XGBoost classifiers trained on the PhiUSIIL dataset
- **MoodFlix**: unsupervised movie recommender using TF-IDF feature learning, cosine similarity scoring, and hybrid ranking logic
- **More on my portfolio**: n8n morning assistant automation, Eurostat gender pay gap dashboard, Titanic survival prediction (team lead on features and training)

ADDITIONAL

- **Certifications**: The Complete Java Certification Course (Udemy); Artificial Intelligence and Machine Learning Course (BITS Pilani); Ethical Hacking and Information Security
- **Languages**: English (full professional), Telugu (native), Swedish (elementary)
- **Awards**: Winner, National Android Developer Challenge (ARK Techno Solutions); National Robotics Championship (IIT Bombay)